# MAT357
# Foundations of Real Analysis

## Class Lecture Notes

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# Contents

**Preface** **ii**

**I Introduction and Preliminaries** **1**

1 Review, Metric, and Normed Vector Spaces 1

2 Metric Space Topology 5

**II Metric Spaces of Functions** **15**

3 Basics(?) 15

4 Arzela-Ascoli Theorem 19

5 Convolutions 21

6 Baire Category Theorem 32

7 Fourier Series 36

**III Measure Theory** **46**

8 Basics 46

# Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

# III Measure Theory

## 8 Basics

riemann integral =(. "im kind of biased in saying this, but im also correct, so". fire.

**Definition 8.1**

for an int'vl $I \subseteq \mathbb{R}$ w endpts $a < b$, define $\ell(I) = b - a$. for $A \subseteq \mathbb{R}$, its **outer measure** $m^*(A)$ is:

$$m^*(A) := \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) : A \subseteq \bigcup_{k=1}^{\infty} I_k,\ I_k \text{ open bdd int'vls} \right\}$$

note $m^*(A) \in [0, \infty]$, i.e. $\mathbb{R}_{\geq 0}$ but can also be infty.

**Proposition 8.2**

i) if $A \subseteq B$, $m^*(A) \leq m^*(B)$

ii) if $E$ ctbl, $m^*(E) = 0$

iii) if $I$ int'vl, then $m^*(I) = \ell(I)$

**Proof.**

i) any cvr of $B$ cvrs $A$.

ii) if $E = \{x_i\}$, let $I_k = (x_k - \varepsilon/2^{k+1}, x_k + \varepsilon/2^{k+1})$. then:

$$m^*(E) \leq \sum \ell(I_k) = \sum \varepsilon/2^k = \varepsilon$$

iii) sps $J = [a, b]$ (renamed). sps $I_k$ open int'vls cvring $J$. $J$ cpt, so take finitely

many; also assume no $I_j \subseteq I_k$. if $I_k = (a_k, b_k)$ and $\{a_i\}, \{b_i\}$ increasing, then:

$$\sum_{j=1}^{k} \ell(I_j) \;=\; \sum_{j=1}^{k} b_j - a_j \;=\; b_k - a_1 + \sum b_j - a_{j+1} \;>\; b_k - a_1 \;\geq\; b - a$$

so $\forall$ cvrs $\{I_j\}$, $\sum \ell(I_j) \geq b - a$ implies $m^*(J) \geq b - a$. OTOH, if $J_\varepsilon = (a - \varepsilon, b + \varepsilon)$, then $m^*(J) \geq \ell(J_\varepsilon) \to b - a$. [i love aboos of notation]

now, if $J = (a, b)$, let $I^- = [a + \varepsilon, b - \varepsilon]$ and $I^+ = [a - \varepsilon, b + \varepsilon]$. clearly, $I^- \subseteq J \subseteq I^+$, so $m^*(I^-) \leq m^*(J) \leq m^*(I^+)$, and it follows.

■

**Lemma 8.3**

if $E \subseteq \mathbb{R}$ and $y \in \mathbb{R}$, then $m^*(E + y) = m^*(E)$.

translate ur cvrs.

**Lemma 8.4**

if $E \subseteq \bigcup_{k=1}^{\infty} E_k$, then $m^*(E) \leq \sum m^*(E_k)$.

**Proof.**

for all $k$, let $\left\{I_j^{(k)}\right\}$ be an int'vl cvr of $E_k$ s.t.:

$$m^*(E_k) \;\leq\; \sum_{j=1}^{\infty} \ell(I_j^{(k)}) \;\leq\; m^*(E_k) + \frac{\varepsilon}{2k}$$

since $\left\{I_j^{(k)}\right\}_{j,k}$ an int'vl cvr of $E$, then:

$$m^*(E) \;\leq\; \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\ell(I_j^{(k)}) \;\leq\; \sum_k m^*(E_k) + \frac{\varepsilon}{2k} \;\leq\; \sum_k m^*(E_k) \;+\; \varepsilon$$

■

**Definition 8.5**

a set $E \subseteq \mathbb{R}$ is **measurable** if $\forall A \subseteq \mathbb{R}$:

$$m^*(A) \;=\; m^*(A \cap E) + m^*(A \cap E^c)$$

we denote by $\mathcal{E}$ the set of measurable sets.

note by the union property, we always have $m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c)$. thus, to show $E$ measurable, we only need to show:

$$m^*(A) \;\geq\; m^*(A \cap E) + m^*(A \cap E^c)$$

note also, non-measurable sets exist; that is, $\exists A, B$ disjoint s.t.:

$$m^*(A \cup B) \;\lneq\; m^*(A) + m^*(B)$$

**Lemma 8.6**

if $m^*(E) = 0$, then $E \in \mathcal{E}$.

**Proof.**

note $m^*(A \cap E) + m^*(A \cap E^c) \;\leq\; m^*(E) + m^*(A) \;=\; m^*(A)$. ■

**Lemma 8.7**

if $E_1, E_2 \in \mathcal{E}$, then $E_1 \cup E_2 \in \mathcal{E}$.

**Proof.**

sps $A \subseteq \mathbb{R}$. then:

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c) \\ &\geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c) \end{aligned}$$

since $A \cap (E_1 \cup E_2) \;=\; (A \cap E_1) \cup (A \cap E_1^c \cap E_2)$. ■

**Definition 8.8**

a coll'n $\mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$ is an **algebra** if:

- $\mathbb{R} \in \mathcal{A}$
- $E \in \mathcal{A} \implies E^c \in \mathcal{A}$
- $E_1 \cup E_2 \in \mathcal{A} \implies E_1 \cup E_2 \in \mathcal{A}$

**Corollary 8.9**

$\mathcal{E}$ is an alg

**Proposition 8.10**

if $C_1, C_2 \in \mathcal{E}$ disj, then $m^*(C_1 \cup C_2) = m^*(C_1) + m^*(C_2)$.

**Proof.**

note $A \cap E_1 \;=\; A \cap E_1 \cap E_2 + A \cap E_1 \cap E_2^c$, so:

$$\begin{aligned} m^*(A \cap (E_1 \cup E_2)) &= m^*(A \cap (E_1 \cup E_2) \cap E_2) + m^*(A \cap (E_1 \cup E_2) \cap E_2^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_2) \end{aligned}$$

■

**Proposition 8.11**

$\mathcal{E}$ closed under ctbl unions.

**Proof.**

let $A_j \in \mathcal{E}$. denote:

$$E_1 \;=\; A_1 \qquad E_j \;=\; A_j \setminus \bigcup_{k=1}^{j-1} A_k$$

then $E_j$ disj and $E = \bigcup E_j$. let $F_n = \bigcup_{j=1}^n E_j$. then:

$$\begin{aligned} m^*(A) &= m^*(A \cap F_n) + m^*(A \cap F_n^c) \\ &\geq m^*(A \cap F_n) + m^*(A \cap E^c) \\ &= \sum_{j=1}^n m^*(A \cap E_j) + m^*(A \cap E^c) \end{aligned}$$

taking $n \to \infty$:

$$m^*(A) \;\geq\; \sum_{j=1}^\infty m^*(A \cap E_j) + m^*(A \cap E^c) \;\geq\; m^*(A \cap E) + m^*(A \cap E^c)$$

■

**Definition 8.12**

a $\sigma$-algebra is an alg closed under ctbl unions.

thus, $\mathcal{E}$ is a $\sigma$-alg. note also that $\mathcal{E}$ closed under ctbl intersections by taking complements.

**Definition 8.13**

we call $\mathcal{E}$ the set of **Lebesgue measurable** sets, and denote $m := m^*|_{\mathcal{E}}$ as the **Lebesgue measure**.

**Lemma 8.14**

$(a, \infty) \in \mathcal{E}$.

**Proof.**

let $A \subseteq \mathbb{R}$; first sps $a \notin A$.

sps $\{I_j\}$ cvrs $A$. let $J_k = I_k \cap (-\infty, a), K_k = I_k \cap (a, \infty)$. note:

$$\sum_j \ell(I_j) \;=\; \sum_j \ell(J_j) + \sum_j \ell(K_j)$$

we also have:

$$A \cap (-\infty, a) \;\subseteq\; \bigcup_j J_j \qquad A \cap (a, \infty) \;\subseteq\; \bigcup_j K_j$$

therefore, we have that:

$$m^*(A \cap (-\infty, a)) + m^*(A \cap (a, \infty)) \;\leq\; \sum_j \ell(J_k) + \sum_j \ell(K_j) \;=\; \sum_j \ell(I_j)$$

and it follows that $m^*(A \cap (-\infty, a)) + m^*(A \cap (a, \infty)) \leq m^*(A)$. ■

Lec 16 - Mar 10 (Week 9)

missed

Lec 17 - Mar 17 (Week 10)

midterm last thurs

**Theorem 8.15**

i) if $A_n \subseteq A_{n+1} \in \mathcal{E}$ then $m\left(\bigcup A_n\right) = \lim_n m(A_n)$

ii) if $m(B_1) < \infty$ and $B_n \supseteq B_{n+1}$ then $m\left(\bigcap B_n\right) = \lim_n m(B_n)$

**Proof.**

i) let $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1} = A_n \cap (A_{n-1})^c \in \mathcal{E}$. then they are disj, so:

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = m\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} m(C_n)$$

$$= \lim_{n\to\infty} \sum_{k=1}^{n} m(C_k) = \lim_{n\to\infty} m\left(\bigcup_{k=1}^{n} C_k\right) = \lim_{n\to\infty} m(A_n)$$

ii) let $A_n = B_1 \setminus B_n$. since $B_n$ is decreasing, $A_n$ is increasing, so

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} m(A_n)$$

note:

$$m(A_n) = m(B_1 \setminus B_n) = m(B_1) - m(B_n)$$

OTOH:

$$\begin{aligned} m\left(\bigcup_{n=1}^{\infty} A_n\right) &= m\left(\bigcup_{n=1}^{\infty} B_1 \setminus B_n\right) \\ &= m\left(B_1 \setminus \bigcap_{n=1}^{\infty} B_n\right) \\ &= m(B_1) - m\left(\bigcap_{n=1}^{\infty} B_n\right) \\ &= \lim_{n\to\infty} m(B_1) - m(B_n) \end{aligned}$$

■

recall the defn of a $\sigma$-alg. given a coll'n of subsets of $\mathbb{R}$ the $\sigma$-alg *generated by* $A$ is the intersection of all $\sigma$-algs containing $A$, denoted $\sigma(A)$.

**Definition 8.16**

a **measure space** $(\Omega, \mathcal{F})$ is a set $\Omega$ and a $\sigma$-alg $\mathcal{F}$ on $\Omega$. given two measure spaces $(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2)$, we say $f : \Omega_1 \longrightarrow \Omega_2$ is **measurable** if $\forall E \in \mathcal{F}_2$, we have that $f^{-1}(E) \in \mathcal{F}_1$.

note the similarity to (topological) continuity, not that the similarity is particularly important though.

**Lemma 8.17**

if $(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2)$ are measure spaces s.t. $\mathcal{F}_2 = \sigma(A)$ for some coll'n $A \subseteq \mathcal{P}(\Omega_2)$, then $f : \Omega_1 \longrightarrow \Omega_2$ is measurable if:

$$f^{-1}(E) \in \mathcal{F}_1$$

for all $E \in A$.

**Proof.**

suffices to check that

$$S := \{E : f^{-1}(E) \in \mathcal{F}_1\}$$

is a $\sigma$-alg containing $A$. indeed, its a $\sigma$-alg essentially bc $\mathcal{F}_1$ is.

$$E_n \in S \implies f^{-1}\left(\bigcup_{n=1}^{\infty} E_n\right) = \bigcup_{n=1}^{\infty} f^{-1}(E_n) \in \mathcal{F}_1 \implies \bigcup_{n=1}^{\infty} E_n \in S$$

other axioms are similar. ■

so why do we care abt this lemma?

**Definition 8.18**

the **Borel $\sigma$-alg** is given as:

$$\mathcal{B}(\mathbb{R}) := \sigma(\text{open subsets of } \mathbb{R})$$

fact: $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{E} \subsetneq 2^{\mathbb{R}}$. the inclusions being strict is tedious and unnecessary to show, and thus skipped.

**Definition 8.19**

we say a fn $f : \mathbb{R} \longrightarrow \mathbb{R}$ is **Lebesgue measurable** if it is measurable as a fn $(\mathbb{R}, \mathcal{E}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. we say it is **Borel measurable** if it is measurable as a fn $(\mathbb{R}, \mathcal{B}(\mathbb{R})) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

**Lemma 8.20**

a fn $f : \mathbb{R} \longrightarrow \mathbb{R}$ is lebesgue (borel) meas if:

$$f^{-1}((c, \infty)) \in \mathcal{E} \qquad (\text{or } \mathcal{B}(\mathbb{R}))$$

for any $c \in \mathbb{Q}$.

**Proof.**

suffices to check $\{\text{open sets}\} \subseteq \sigma(\{(c, \infty)\}) =: \mathcal{F}$ for $c \in \mathbb{Q}$.

i) check $(a, b) \in \mathcal{F}$. indeed, $(a, b) = (a, \infty) \setminus [b, \infty)$, but $[b, \infty) = \bigcap (q, \infty)$ for $q \in \mathbb{Q} \cap (-\infty, b)$.

ii) any open set in $\mathbb{R}$ is a ctbl union of open intvls w/ rational endpts.

■

**Corollary 8.21**

any cts or monotonic $f : \mathbb{R} \longrightarrow \mathbb{R}$ is borel measurable.

**Proposition 8.22**

sps $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ are lebesgue (borel) meas. the following are as well:

i) $|f|, \alpha f + \beta$ for real $\alpha, \beta$

ii) $(f)_+, (f)_-$ where $(x)_*$ is the $+$ve or $-$ve parts of $x$

**Proof.**

these are compositions of $f$ w/ a cts fn, hence borel meas. easy to see that the composition of a BM fn w/ a LM fn is LM.

■